Doping Stability Report -- Y in LiCoO2 at 350 C

Generated: 2026-05-31 15:12   Host: LiCoO2   Dopant: Y

Executive Summary

Y preferentially occupies the Co layer. The formation energy difference between the two layers is +7.831 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau slope is -0.00051 A^2/ps (stable), compared to -0.00111 A^2/ps on the Li site. The Co-site result is isovalent (no charge mismatch), so the formation energy is fully reliable. The Li-site E_f is approximate due to a charge surplus of +2 that cannot be modelled in CHGNet.

Site	E_f (eV)	Charge	Compensation	MSD pass	Coord pass	Verdict
Co	-4.209	+0	No	PASS	PASS	STABLE
Li	+3.622	+2	Yes	PASS	FAIL	STRUCTURAL COLLAPSE

Y at the Co layer -- STABLE

Charge situation

Y replaces Co with no change in oxidation state (isovalent substitution, charge mismatch = 0). No charge compensation is needed. The formation energy is fully reliable.

Formation Energy (thermodynamic stability)

Formation energy E_f	**-4.2086 eV**	PASS	*< 1.0 eV to pass*

E_f = -4.209 eV. This value is exceptionally large and negative, indicating that Y incorporation is strongly thermodynamically driven at the Co site.

Molecular Dynamics Stability (350 C, 25 ps)

MSD slope (late-time)	**-0.00051 A^2/ps**	PASS	*< 0.005 to pass (plateau = stable)*
Final MSD	**0.023 A^2**	----	
Max displacement	**0.574 A**	----	
Coordination (min/mean/max)	**5 / 6.0 / 7**	PASS	*relaxed = 6, tolerance +/-1*
Mean bond length (MD vs relaxed)	**2.150 A (relaxed: 2.1**	----	
Lattice volume change	**+0.00 %**	PASS	*< +/-5% to pass*

Late-time MSD slope = -0.00051 A^2/ps. The negative slope is a flat plateau -- essentially no migration detected. Final MSD = 0.023 A^2; maximum displacement from starting position = 0.57 A.

Coordination fluctuated between 5 and 7 (mean 6.0), within the +/-1 tolerance of the relaxed value (6). This is consistent with normal thermal vibration. Mean nearest-neighbour distance during MD = 2.150 A.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: STABLE

Y is thermodynamically favorable and kinetically stable on the Co site at 350 C. All four stability criteria pass: formation energy is favorable, the dopant does not migrate during MD, its coordination environment is preserved, and the host lattice volume is unchanged.

Y at the Li layer -- STRUCTURAL COLLAPSE

Charge situation

Y replaces Li with a charge surplus of +2. CHGNet cannot model this compensation (would require Co oxidation or electron addition). The formation energy below is approximate.

Formation Energy (thermodynamic stability)

Formation energy E_f	**+3.6221 eV**	FAIL	*< 1.0 eV to pass*

E_f = +3.622 eV (approximate -- charge surplus not modelled in CHGNet). This value is large and positive, indicating that Y incorporation is thermodynamically unfavorable under equilibrium conditions at the Li site.

Molecular Dynamics Stability (350 C, 25 ps)

MSD slope (late-time)	**-0.00111 A^2/ps**	PASS	*< 0.005 to pass (plateau = stable)*
Final MSD	**0.200 A^2**	----	
Max displacement	**1.111 A**	----	
Coordination (min/mean/max)	**5 / 6.0 / 8**	FAIL	*relaxed = 6, tolerance +/-1*
Mean bond length (MD vs relaxed)	**2.205 A (relaxed: 2.1**	----	
Lattice volume change	**+0.00 %**	PASS	*< +/-5% to pass*

Late-time MSD slope = -0.00111 A^2/ps. The negative slope is a flat plateau -- essentially no migration detected. Final MSD = 0.200 A^2; maximum displacement from starting position = 1.11 A.

Coordination fluctuated between 5 and 8 (mean 6.0), deviating by more than +/-1 from the relaxed value (6). This may indicate structural distortion, or that the bond-length cutoff is too close to the actual bond length -- check the coordination_cutoff_A setting. Mean nearest-neighbour distance during MD = 2.205 A.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: STRUCTURAL COLLAPSE

The coordination criterion failed while MSD and volume both pass, suggesting a possible cutoff artifact rather than a true structural collapse. The Y atom barely moves (MSD = 0.200 A^2) but the coordination count fluctuates because the bond length at the Li site is close to the analysis cutoff. Consider increasing coordination_cutoff_A and re-running analysis.

Site Preference Comparison

The Co site is preferred by 7.831 eV over the Li site. Formation energy: -4.209 eV (Co) vs +3.622 eV (Li).

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Caveats and Limitations

? **CHGNet is charge-neutral.**

The potential does not model oxidation states, polaron formation, or electron/hole localisation. Formation energies for aliovalent dopants (charge mismatch != 0) are approximate even when structural compensation defects are included.

? **No Freysoldt/Kumagai correction.**

Electrostatic finite-size corrections for charged defects require the host dielectric tensor and the DFT electrostatic potential, neither of which is available from a neural-network potential. For publication-quality E_f of charged defects, follow up with DFT using doped or pymatgen-analysis-defects.

? **Metal-rich reference state.**

Chemical potentials are computed from elemental ground-state structures. Synthesis under oxide-rich or other conditions shifts all E_f values by additive constants; the relative ordering of sites is unaffected.

? **MD timescales are short (25 ps default).**

This is sufficient to assess local site stability and rapid migration, but cannot resolve slow diffusion mechanisms. A non-plateauing MSD confirms instability; a plateau does not guarantee stability on longer timescales.

? **Energy rankings are more trustworthy than absolute values.**

CHGNet is a universal MLIP trained on r2SCAN DFT. Formation-energy errors for transition-metal oxides are typically 50-150 meV/atom. Use relative orderings (which site is preferred?) as the primary result.